Cell organelles 2

Dr.Iman Nabil

DrImanNabil


Why?
tell me
why!



Cell organelles II

1- Golgi apparatus: structure& functions.

2- Endosomes: types, structure & functions.

3- Lysosomes: structure& function.

Dr Iman Nabil

Golgi apparatus

Dr Imam Sahil

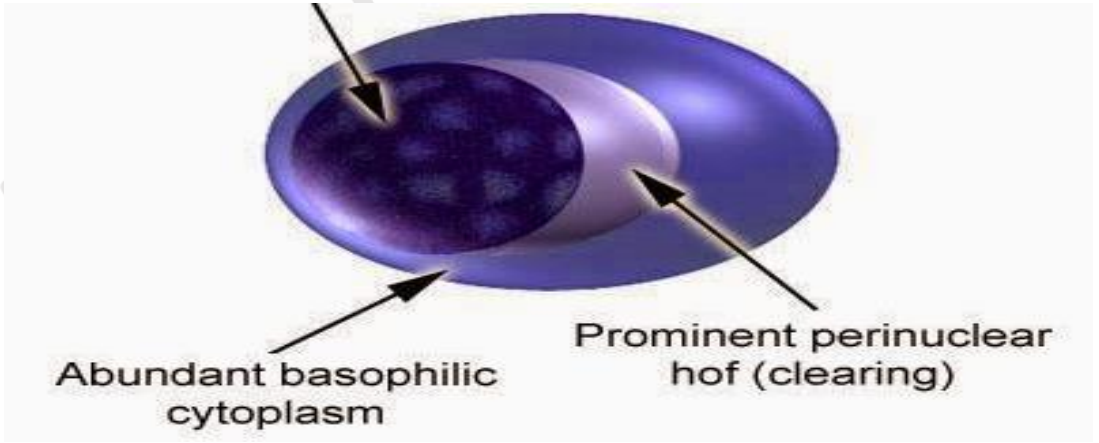
Golgi apparatus

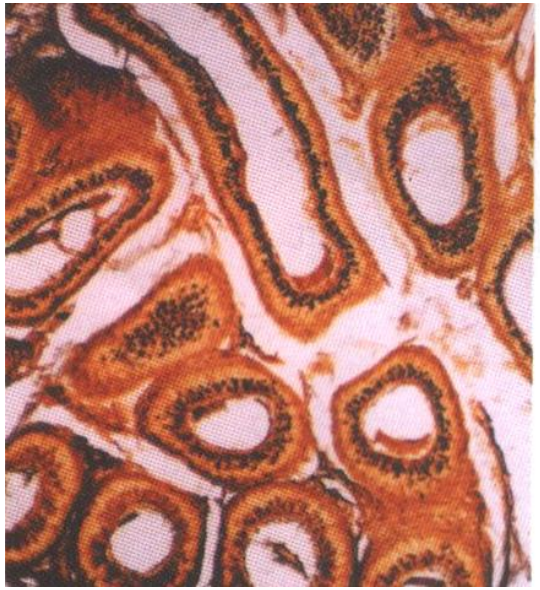
- **LM :**

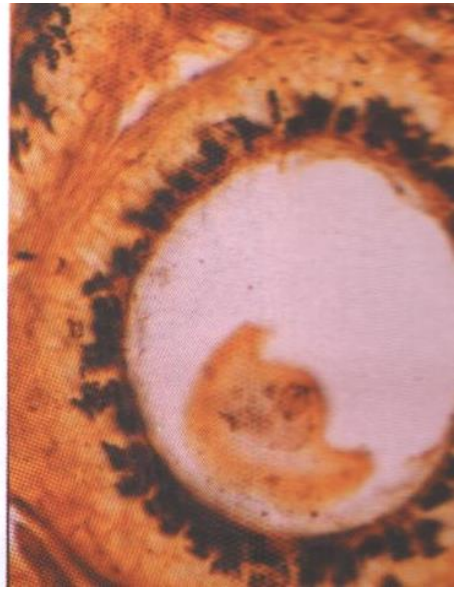
1-It is *not stained* with H&E in routine histological sections.

2-H&E-stained sections of the cells synthesizing protein, its site could be seen as *negative Golgi image* e.g. plasma cells.

3-It can be seen in histological sections stained with *silver stain*.







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Golgi apparatus

- **EM :**

1-It consists of saccules or cisternae called the *Golgi stacks*.

2-Each stack consists of 4- 6 *cisternae* (flattened, curved, membrane-bounded, slightly expanded at the ends).

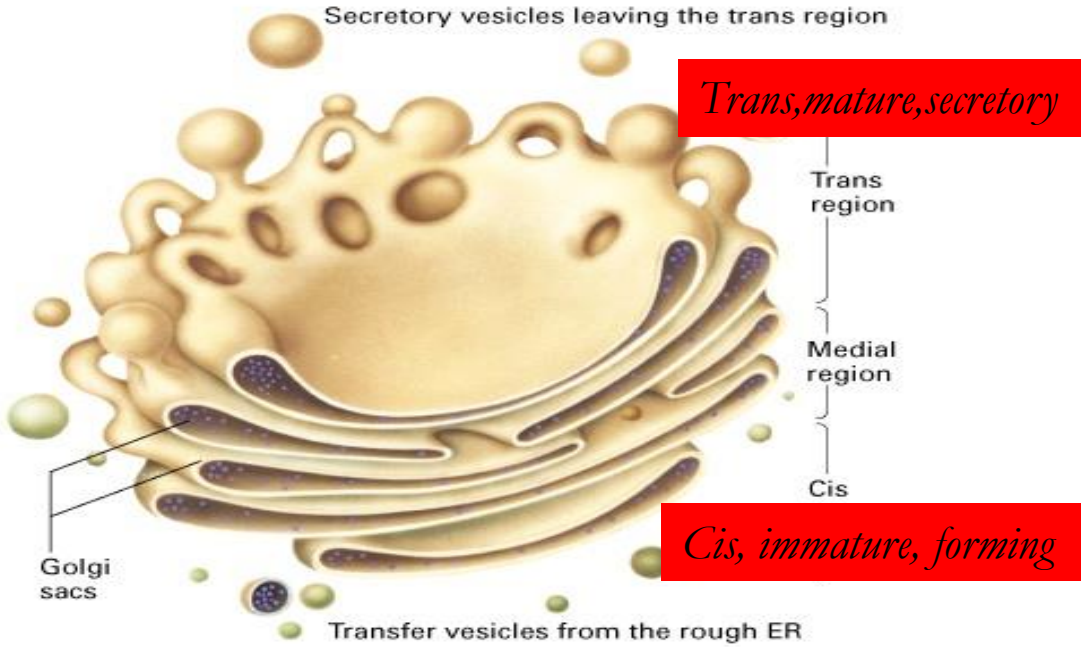
3-The Golgi stack is *cup-shaped* with a convex surface and a concave surface:

a. ***The cis face*** (***forming face***; ***immature face)***:is convex in shape and lies near to the rER.

b. ***The trans face*** (***secretory face; mature face)*** is concave in shape.

c. ***The medial compartment*** .

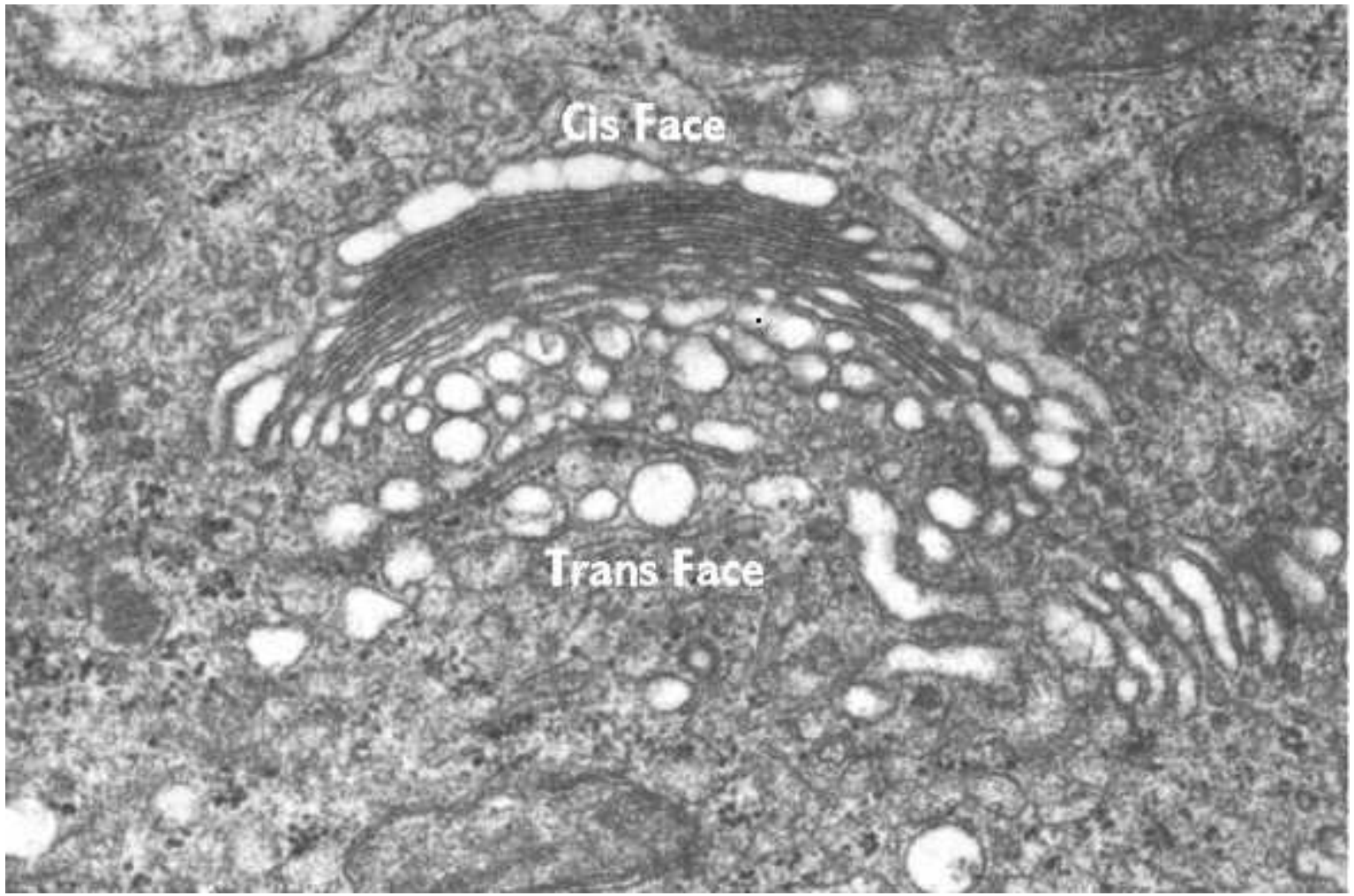




Trans,mature,secretory

Cis, immature, forming

Dr Iman N


Cis Face
Trans Face

Golgi apparatus

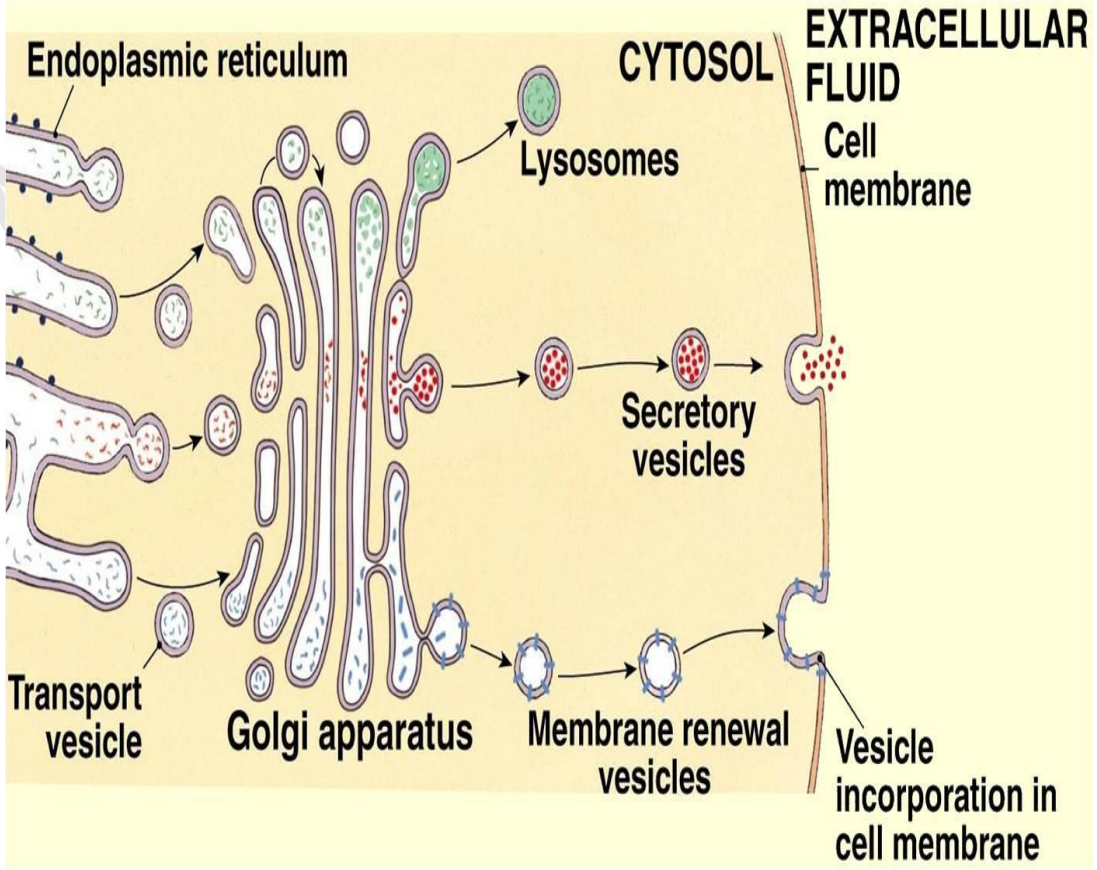
- **Functions:**

1- *Post-translational modifications* of proteins e.g. addition of sugars.

2- *Packaging* of different proteins in membrane bounded vesicles.

3- *Sorting and targeting of vesicles* to the right destination:

- Formation of lysosomes.
- Formation of secretory granules for exocytosis.
- Membrane recycling.



Sites: Protein synthesizing and secretory cells

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Endosomes

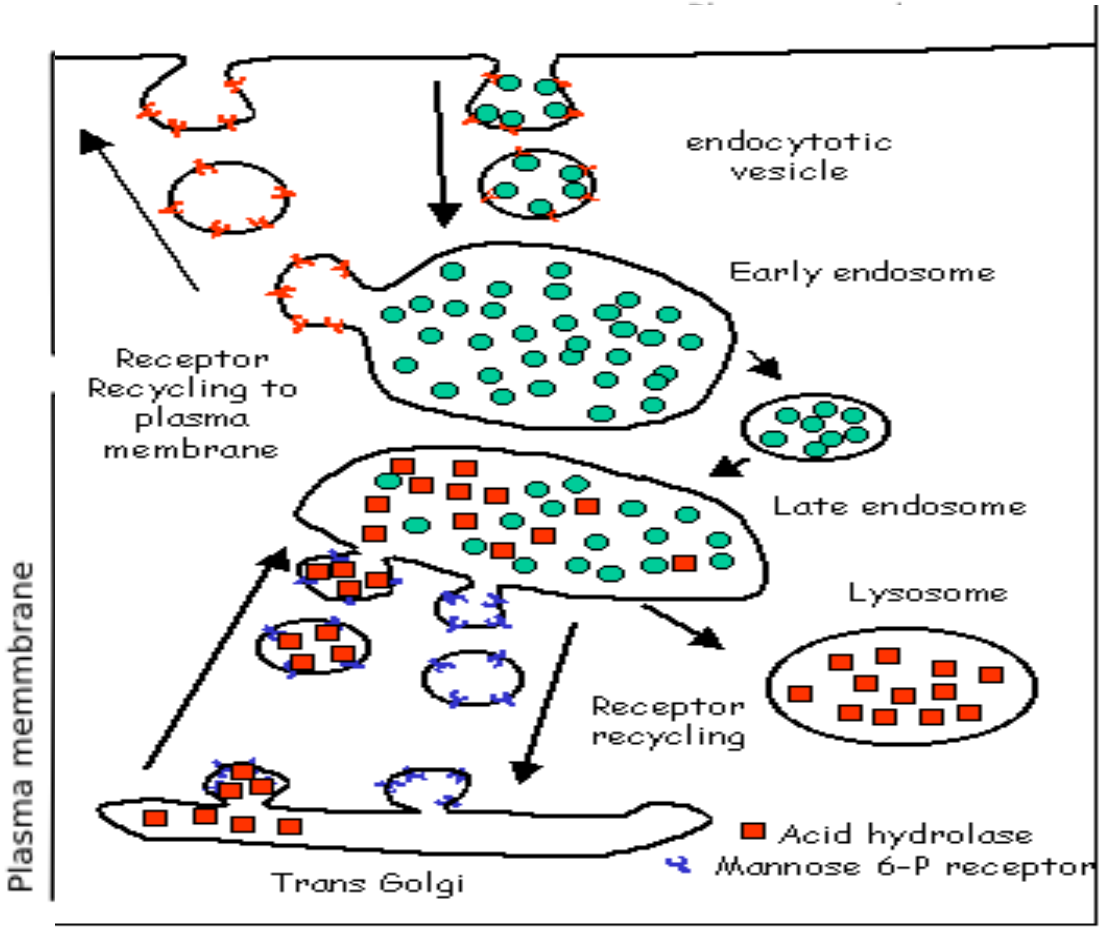
Dr Imam Mobil

Endosomes

- **Definition**: are system of vesicles and tubules involved in the endocytotic pathway.
- **Types:**

A-Early endosomes:

- Site: at the *periphery of the cell* as a part in the pathway of the receptor-mediated endocytosis.
- Content: the receptors-ligands complex.
- The membrane of the endosomes pumps H^+ ions into its interior ⟶ lowers the pH of endosomes to less than 6 ⟶ *uncoupling* of the receptors and the ligands.
- The receptors recycle to the cell membrane and the ligands move to the late endosomes.



Endosomes

B-Late endosomes:

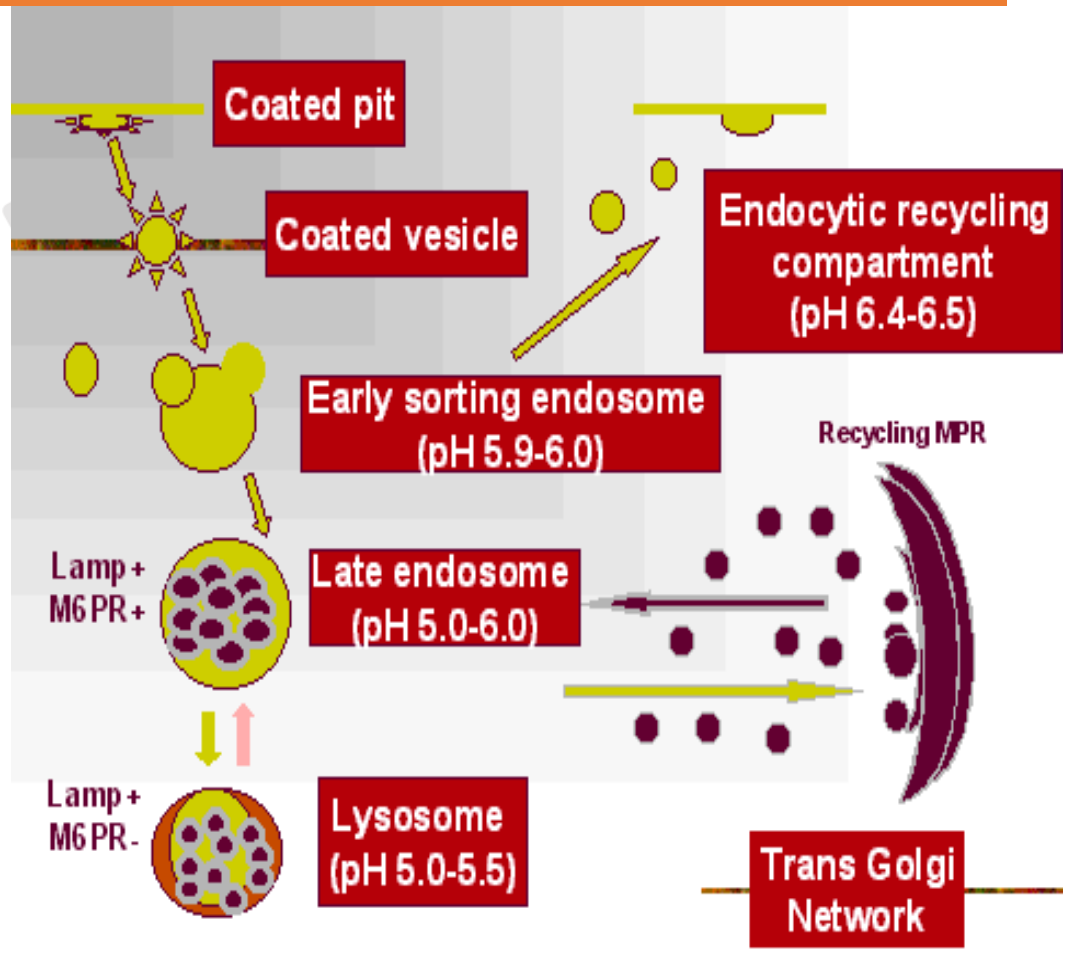
- Site: *deep* within the cytoplasm near the Golgi complex.
- They receive:

1- *the ligands* from early endosomes.

2- *clathrin- coated vesicles* containing lysosomal enzymes from Golgi complex.

- PH: 5.5. (the enzymes become active at the acidic pH of the late endosome).

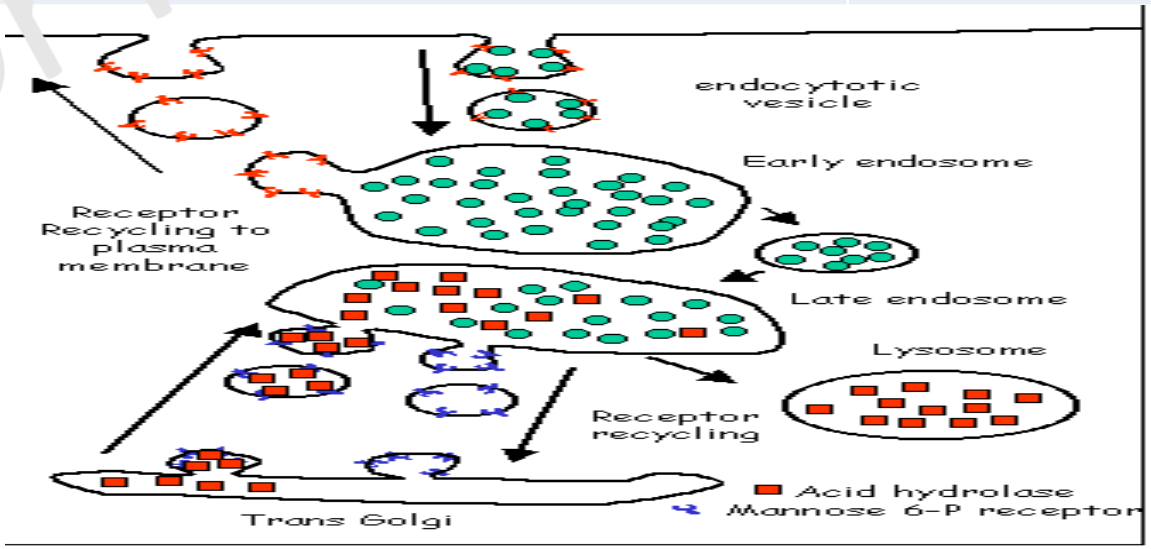
- *The lysosomal enzymes in the late endosome begin to degrade the ligands accompanied by further decrease in the internal pH which then "mature" to form lysosomes.*



Dr Iman

Types of endosomes	Early endosome	Late endosome
1-Site	Periphery of cytoplasm.	Deep in cytoplasm , near Golgi.
2- Content	Receptor ligand complex.	1- Ligands from early endosome. 2- Lysosomal enzymes from Golgi.
3-Function	Uncoupling of the receptor from ligand.	Lysosomal enzymes begin to derade ligands, then the late endosome matures to lysosome.
4- pH	Less than 6.	5.5.

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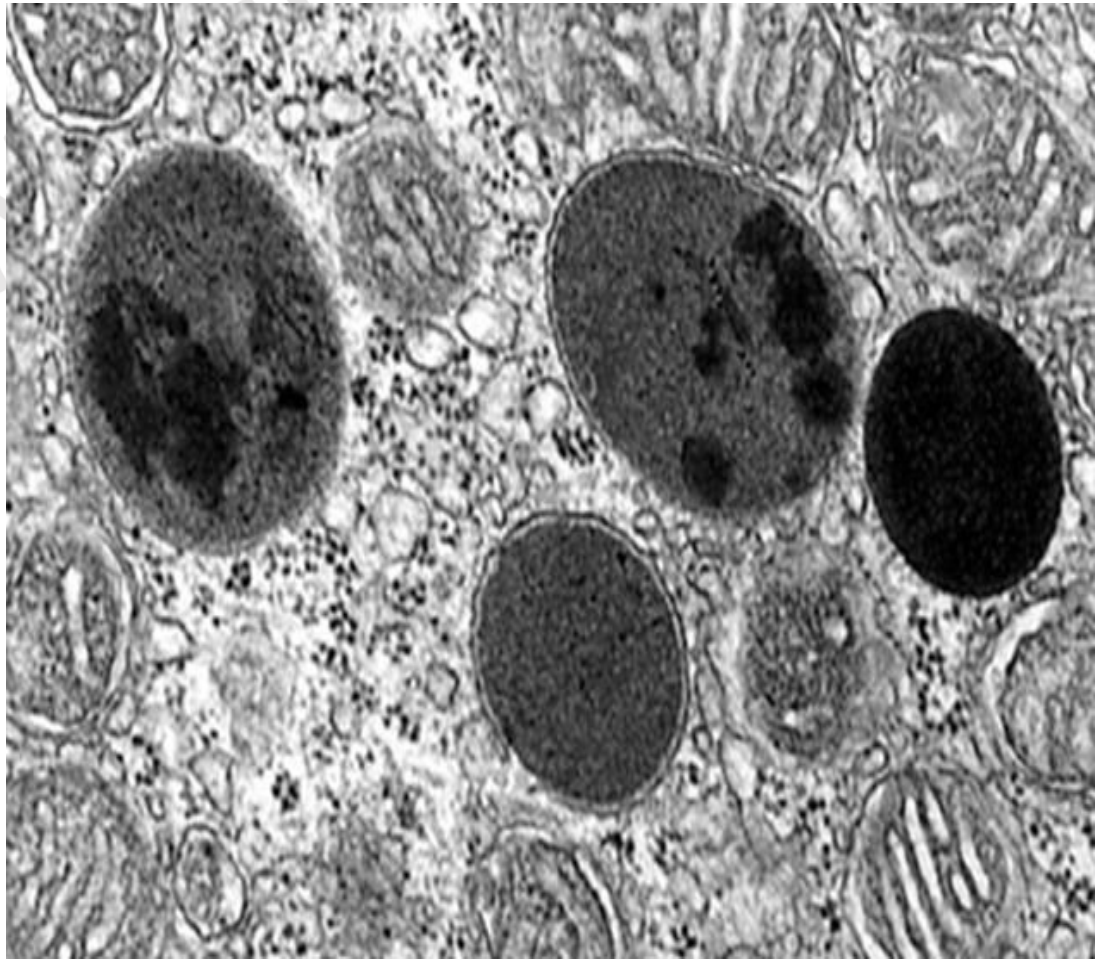


Lysosomes

Dr Imaan Bashil

Lysosomes

- Definition: membrane-bounded organelle containing about 40 types of *acid hydrolytic digestive enzymes* (proteases, nucleases, phosphatases, and lipases).
- Function: They are responsible for *intracellular digestion* of different materials.
- Synthesis: by a *gradual maturation process* as by fusion of the clathrin- coated vesicles coming from Golgi complex with late endosomes.



Dr Iman A

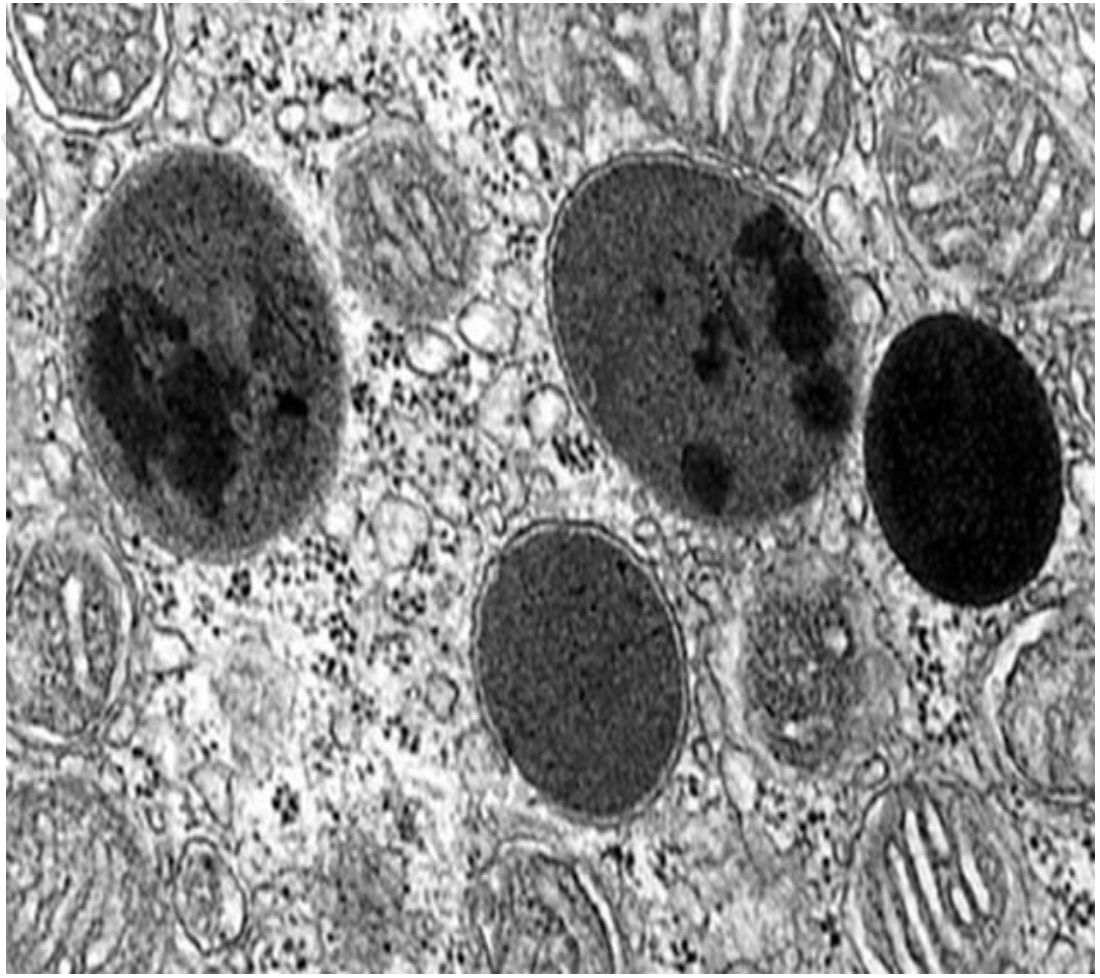
Lysosomes

- Lysosome has a *surrounding membrane with unique phospholipids* that prevents:

1- the leak out of the enzymes to the cytoplasm.

2- protects the membrane from hydrolysis by its own enzymes.

- If a lysosome leaks its contents, the released enzymes would be inactive because of neutral pH of the cytoplasm.

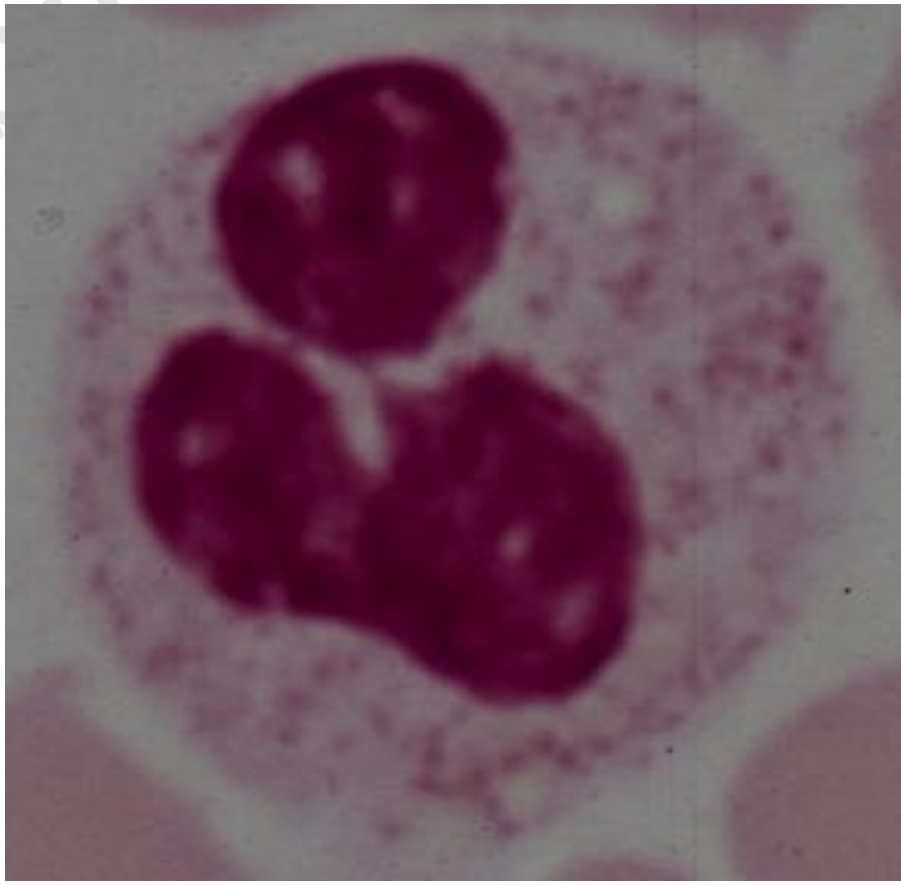


Dr Iman A

Lysosomes

- **LM :**

They can be recognized by several histochemical methods used to demonstrate the lysosomal enzymes.



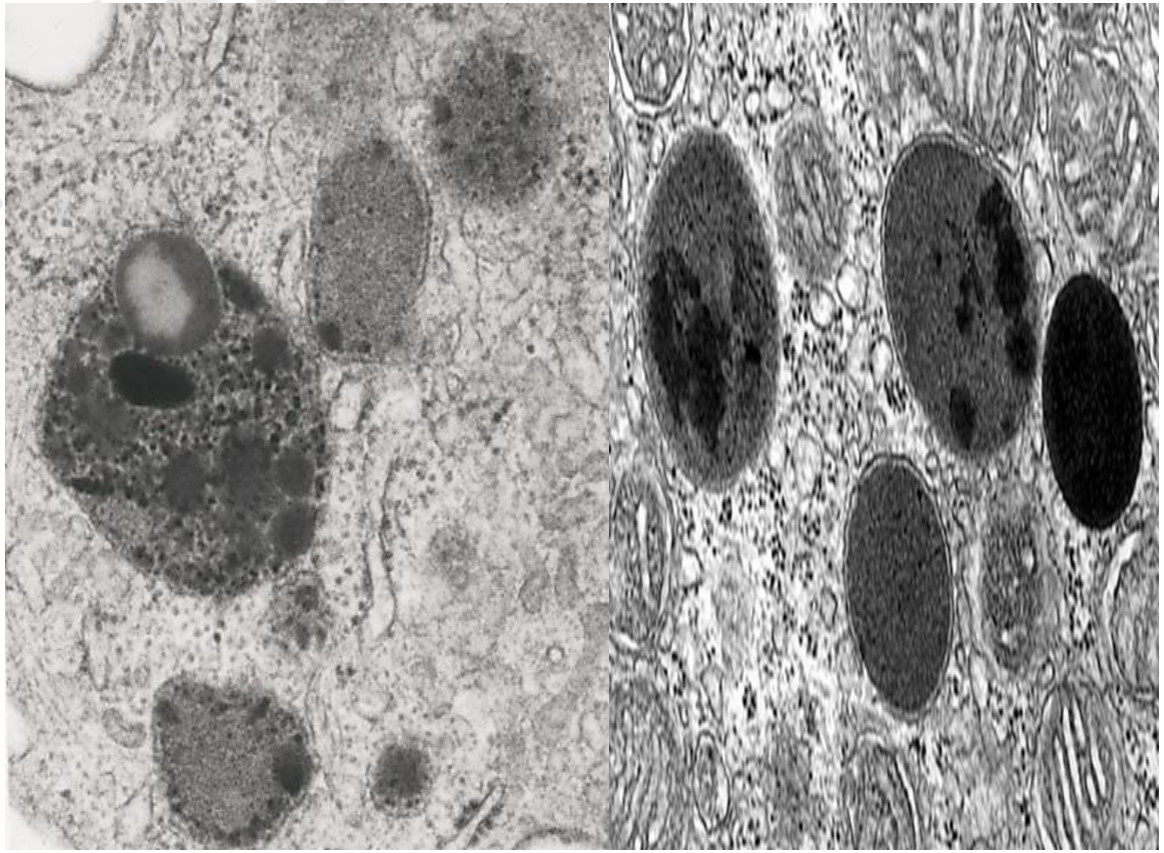
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Lysosomes

- **EM:**

Lysosomes are *heterogeneous* in shape and the appearance of their interior. Some are electron dense, others show electron lucent areas.

- **Site**: are abundant in phagocytic cells.



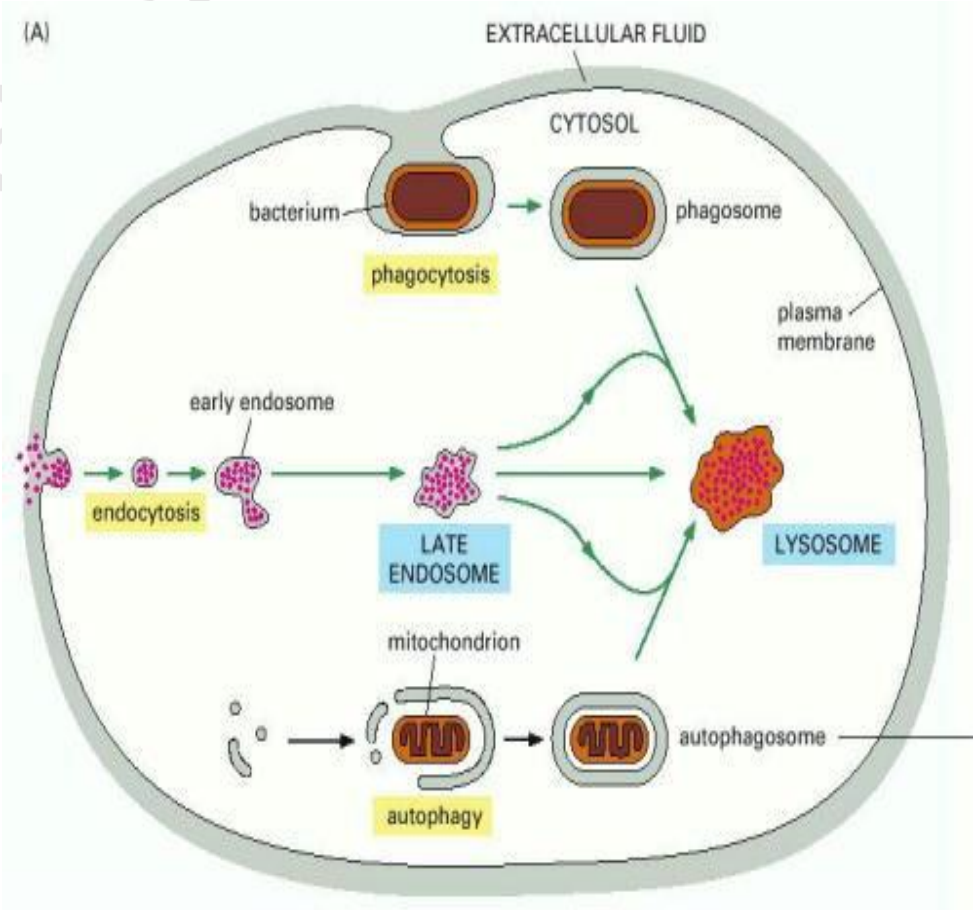
Dr Iman Habil

Pathways for intracellular digestion by lysosomes

1-Extracellular small particles: internalized by pinocytosis and receptor-mediated *endocytosis* ⟶ *early endosome* ⟶ *late endosome* where the endocytosed materials are degraded by the lysosomal hydrolases.

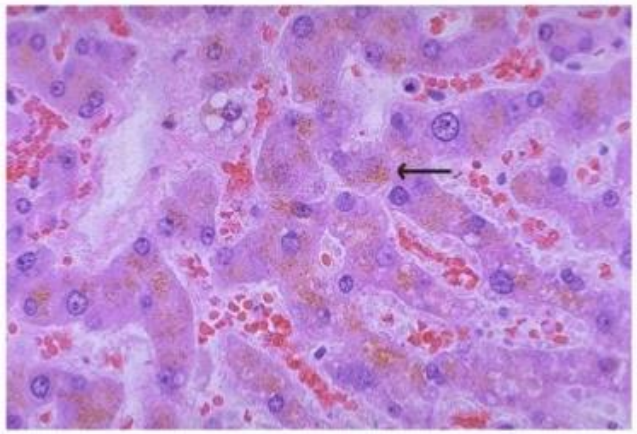
2-Extracellular large particles: are engulfed in the process of *phagocytosis* ⟶ forms a *phagosome* ⟶ fuses with a late endosome.

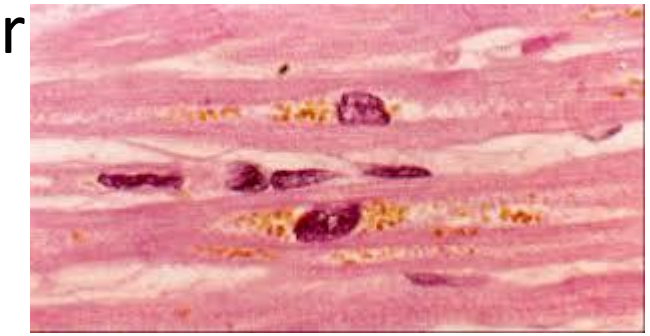
3- Intracellular particles: are removed by a process called *autophagy* ⟶ the enclosure of this organelle by membranes from sER ⟶ forms an *autophagosome* ⟶ fuses with a late endosome.

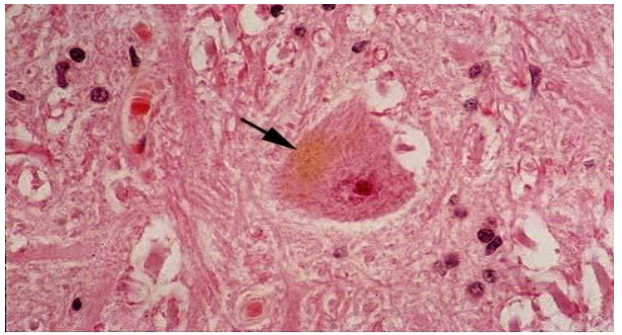


Pathways for intracellular digestion by lysosomes

- The hydrolytic enzymes digest most of the content of the lysosomes. Any indigestible substances remain in lysosomes forming *residual bodies*.
- In long-lived cells, accumulated residual bodies indicate cellular aging and are called *lipofuscin pigments.*







Dr Iman N

Which pathway for intracellular protein synthesis is correct?

1- sER……Golgi……secretory vesicles.

2- rER……Ribosomes……secretory vesicles.

3- Ribosomes……Golgi……rER.

4- rER……Golgi……Secretory vesicles.

Dr Iman Nabil

Which pathway for intracellular protein synthesis is correct?

1- sER……Golgi……secretory vesicles.

2- rER……Ribosomes……secretory vesicles.

3- Ribosomes……Golgi……rER.

4- rER……Golgi……Secretory vesicles.

Dr Iman Nabil


THANK
YOU


Thank You